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PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space

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Abstract

Few prior works study deep learning on point sets. PointNet [20] is a pioneer in this direction. However, by design PointNet does not capture local structures induced by the metric space points live in, limiting its ability to recognize fine-grained patterns and generalizability to complex scenes. In this work, we introduce a hierarchical neural network that applies PointNet recursively on a nested partitioning of the input point set. By exploiting metric space distances, our network is able to learn local features with increasing contextual scales. With further observation that point sets are usually sampled with varying densities, which results in greatly decreased performance for networks trained on uniform densities, we propose novel set learning layers to adaptively combine features from multiple scales. Experiments show that our network called PointNet++ is able to learn deep point set features efficiently and robustly. In particular, results significantly better than state-of-the-art have been obtained on challenging benchmarks of 3D point clouds.

1 Introduction

We are interested in analyzing geometric point sets which are collections of points in a Euclidean space. A particularly important type of geometric point set is point cloud captured by 3D scanners, e.g., from appropriately equipped autonomous vehicles. As a set, such data has to be invariant to permutations of its members. In addition, the distance metric defines local neighborhoods that may exhibit different properties. For example, the density and other attributes of points may not be uniform across different locations — in 3D scanning the density variability can come from perspective effects, radial density variations, motion, etc.

Few prior works study deep learning on point sets. PointNet [20] is a pioneering effort that directly processes point sets. The basic idea of PointNet is to learn a spatial encoding of each point and then aggregate all individual point features to a global point cloud signature. By its design, PointNet does not capture local structure induced by the metric. However, exploiting local structure has proven to be important for the success of convolutional architectures. A CNN takes data defined on regular grids as the input and is able to progressively capture features at increasingly larger scales along a multi-resolution hierarchy. At lower levels neurons have smaller receptive fields whereas at higher levels they have larger receptive fields. The ability to abstract local patterns along the hierarchy allows better generalizability to unseen cases.

We introduce a hierarchical neural network, named as PointNet++, to process a set of points sampled in a metric space in a hierarchical fashion. The general idea of PointNet++ is simple. We first partition the set of points into overlapping local regions by the distance metric of the underlying space. Similar to CNNs, we extract local features capturing fine geometric structures from small neighborhoods; such local features are further grouped into larger units and processed to produce higher level features. This process is repeated until we obtain the features of the whole point set.

The design of PointNet++ has to address two issues: how to generate the partitioning of the point set, and how to abstract sets of points or local features through a local feature learner. The two issues

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PointNet++：度量空间中点集的深度层次特征学习

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摘要

以往很少有工作研究点集上的深度学习。PointNet [20] 是该方向的先驱。然而，PointNet 在设计上并未捕捉点集所在度量空间所引发的局部结构，这限制了其识别细粒度模式的能力以及对复杂场景的泛化能力。在本工作中，我们引入了一种层次化神经网络，该网络在输入点集的嵌套划分上递归地应用 PointNet。通过利用度量空间中的距离，我们的网络能够学习具有递增上下文尺度的局部特征。进一步观察到点集通常以不同密度采样，这导致在均匀密度上训练的网络性能大幅下降，我们提出了新颖的集合学习层，以自适应地组合来自多个尺度的特征。实验表明，我们称之为 PointNet++ 的网络能够高效且稳健地学习深度点集特征。特别是，在具有挑战性的三维点云基准上，我们取得了显著优于现有技术水平的结果。

1 引言

我们感兴趣的是分析几何点集，即欧几里得空间中点的集合。一类特别重要的几何点集是由3D扫描仪捕获的点云，例如来自配备适当设备的自动驾驶车辆。作为集合，此类数据必须对其成员的排列具有不变性。此外，距离度量定义了可能表现出不同属性的局部邻域。例如，点的密度和其他属性在不同位置可能并不均匀——在3D扫描中，密度变化可能来自透视效应、径向密度变化、运动等。

先前很少有工作研究点集上的深度学习。PointNet [20] 是直接处理点集的开创性工作。PointNet 的基本思想是学习每个点的空间编码，然后将所有单个点特征聚合为全局点云签名。就其设计而言，PointNet 不捕获由度量引起的局部结构。然而，利用局部结构已被证明对卷积架构的成功至关重要。CNN 以规则网格上定义的数据作为输入，并能够沿多分辨率层级逐步捕获越来越大规模的特征。在较低层级，神经元具有较小的感受野，而在较高层级，它们具有较大的感受野。沿层级抽象局部模式的能力使得对未见情况具有更好的泛化性。

我们提出了一种名为PointNet++的层次化神经网络，以层次化方式处理在度量空间中采样的点集。PointNet++的基本思想很简单。我们首先根据底层空间的距离度量，将点集划分为重叠的局部区域。与CNN类似，我们从小的邻域中提取捕获精细几何结构的局部特征；这些局部特征进一步被分组为更大的单元并进行处理，以产生更高层次的特征。这一过程重复进行，直到获得整个点集的特征。

PointNet++的设计需要解决两个问题：如何生成点集的分割，以及如何通过局部特征学习器抽象点集或局部特征。这两个问题

are correlated because the partitioning of the point set has to produce common structures across partitions, so that weights of local feature learners can be shared, as in the convolutional setting. We choose our local feature learner to be PointNet. As demonstrated in that work, PointNet is an effective architecture to process an unordered set of points for semantic feature extraction. In addition, this architecture is robust to input data corruption. As a basic building block, PointNet abstracts sets of local points or features into higher level representations. In this view, PointNet++ applies PointNet recursively on a nested partitioning of the input set.

One issue that still remains is how to generate overlapping partitioning of a point set. Each partition is defined as a neighborhood ball in the underlying Euclidean space, whose parameters include centroid location and scale. To evenly cover the whole set, the centroids are selected among input point set by a farthest point sampling (FPS) algorithm. Compared with volumetric CNNs that scan the space with fixed strides, our local receptive fields are dependent on both the input data and the metric, and thus more efficient and effective.

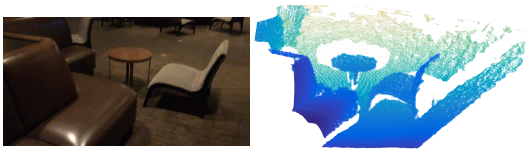


Figure 1: Visualization of a scan captured from a Structure Sensor (left: RGB; right: point cloud).

Deciding the appropriate scale of local neighborhood balls, however, is a more challenging yet intriguing problem, due to the entanglement of feature scale and non-uniformity of input point set. We assume that the input point set may have variable density at different areas, which is quite common in real data such as Structure Sensor scanning [18] (see Fig. 1). Our input point set is thus very different from CNN inputs which can be viewed as data defined on regular grids with uniform constant density. In CNNs, the counterpart to local partition scale is the size of kernels. [25] shows that using smaller kernels helps to improve the ability of CNNs. Our experiments on point set data, however, give counter evidence to this rule. Small neighborhood may consist of too few points due to sampling deficiency, which might be insufficient to allow PointNets to capture patterns robustly.

A significant contribution of our paper is that PointNet++ leverages neighborhoods at multiple scales to achieve both robustness and detail capture. Assisted with random input dropout during training, the network learns to adaptively weight patterns detected at different scales and combine multi-scale features according to the input data. Experiments show that our PointNet++ is able to process point sets efficiently and robustly. In particular, results that are significantly better than state-of-the-art have been obtained on challenging benchmarks of 3D point clouds.

2 Problem Statement

Suppose that $\mathcal{X} = (M, d)$ is a discrete metric space whose metric is inherited from a Euclidean space $\mathbb{R}^n$, where $M \subseteq \mathbb{R}^n$ is the set of points and d is the distance metric. In addition, the density of M in the ambient Euclidean space may not be uniform everywhere. We are interested in learning set functions f that take such $\mathcal{X}$ as the input (along with additional features for each point) and produce information of semantic interest regrading $\mathcal{X}$. In practice, such f can be classification function that assigns a label to $\mathcal{X}$ or a segmentation function that assigns a per point label to each member of M.

3 Method

Our work can be viewed as an extension of PointNet [20] with added hierarchical structure. We first review PointNet (Sec. 3.1) and then introduce a basic extension of PointNet with hierarchical structure (Sec. 3.2). Finally, we propose our PointNet++ that is able to robustly learn features even in non-uniformly sampled point sets (Sec. 3.3).

3.1 Review of PointNet [20]: A Universal Continuous Set Function Approximator

Given an unordered point set $\{x_1, x_2, ..., x_n\}$ with $x_i \in \mathbb{R}^d$, one can define a set function $f : \mathcal{X} \to \mathbb{R}$ that maps a set of points to a vector:

$$f(x_1, x_2, ..., x_n) = \gamma \left(\underset{i=1,...,n}{\text{MAX}} \{h(x_i)\} \right) \quad (1)$$

这些点集的分割必须产生跨分割的公共结构，因此局部特征学习器的权重可以共享，正如卷积设置中的情况一样，所以它们是相关的。我们选择PointNet作为局部特征学习器。如该工作所示，PointNet是处理无序点集以进行语义特征提取的有效架构。此外，该架构对输入数据损坏具有鲁棒性。作为基本构建模块，PointNet将局部点集或特征集抽象为更高级别的表示。从这个角度看，PointNet++在输入集的嵌套分割上递归地应用PointNet。

仍待解决的一个问题是如何生成点集的交叠划分。每个划分被定义为底层欧氏空间中的一个邻域球，其参数包括质心位置和尺度。为均匀覆盖整个集合，质心通过最远点采样（FPS）算法从输入点集中选取。与以固定步长扫描空间的体素CNN相比，我们的局部感受野同时依赖于输入数据和度量，因而更加高效且有效。

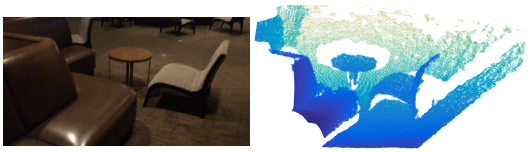


图1：从结构传感器捕获的扫描可视化（左：RGB；右：点云）。

然而，确定局部邻域球的合适尺度是一个更具挑战性且引人入胜的问题，原因在于特征尺度与输入点集非均匀性之间的纠缠。我们假设输入点集在不同区域可能具有可变的密度，这在真实数据中相当常见，例如结构传感器扫描[18]（见图1）。因此，我们的输入点集与CNN输入截然不同，后者可以被视为定义在具有均匀恒定密度的规则网格上的数据。在CNN中，局部划分尺度的对应物是核的大小。[25]表明，使用较小的核有助于提升CNN的能力。然而，我们在点集数据上的实验为这一规则提供了反例。由于采样不足，小的邻域可能包含过少的点，这可能不足以让PointNets稳健地捕获模式。

我们论文的一个显著贡献在于，PointNet++利用多尺度邻域同时实现了鲁棒性和细节捕捉。在训练期间借助随机输入丢弃的辅助，网络学会根据输入数据自适应地加权不同尺度下检测到的模式，并组合多尺度特征。实验表明，我们的PointNet++能够高效且鲁棒地处理点集。特别是，在具有挑战性的3D点云基准上取得了显著优于现有技术水平的结果。

2 问题陈述

假设 $\mathcal{X} = (M, d)$ 是一个离散度量空间，其度量继承自欧几里得空间 $\mathbb{R}^n$，其中 $M \subseteq \mathbb{R}^n$ 是点集，d 是距离度量。此外，M 在周围欧几里得空间中的密度可能并非处处均匀。我们关注于学习集合函数 f，该函数以这样的 $\mathcal{X}$ 作为输入（以及每个点的附加特征），并生成关于 $\mathcal{X}$ 的语义相关信息。在实践中，这样的 f 可以是分类函数，为 $\mathcal{X}$ 分配标签，也可以是分割函数，为 M 的每个成员分配逐点标签。

3 方法

我们的工作可视为PointNet [20]的一种扩展，增加了层次结构。我们首先回顾PointNet（第3.1节），然后介绍PointNet的一种带有层次结构的基本扩展（第3.2节）。最后，我们提出我们的PointNet++，它能够在非均匀采样的点集中稳健地学习特征（第3.3节）。

3.1 PointNet [20] 综述：一种通用的连续集合函数逼近器

给定一个无序点集$\{x_1, x_2, ..., x_n\}$，其中$x_i \in \mathbb{R}^d$，可以定义一个集合函数f：$\mathcal{X} \to \mathbb{R}$，该函数将点集映射为一个向量：

$$f(x_1, x_2, ..., x_n) = \gamma \left(\underset{i=1,...,n}{\text{MAX}} \{h(x_i)\} \right) \tag{1}$$

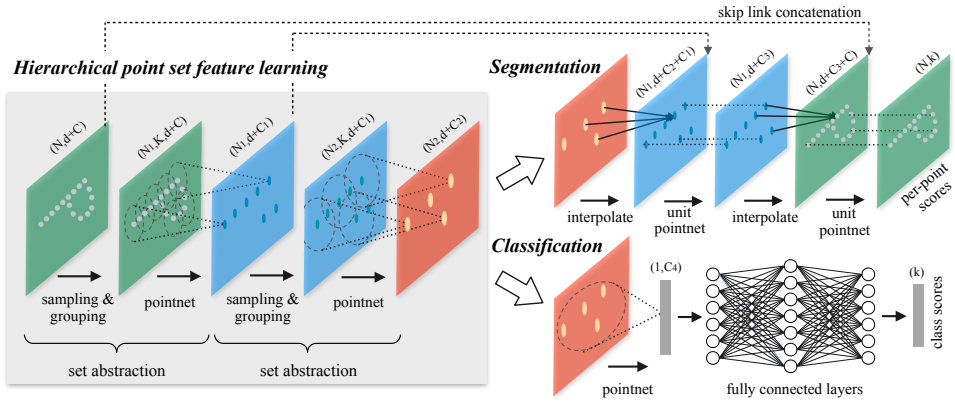


Figure 2: Illustration of our hierarchical feature learning architecture and its application for set segmentation and classification using points in 2D Euclidean space as an example. Single scale point grouping is visualized here. For details on density adaptive grouping, see Fig. 3

where γ and h are usually multi-layer perceptron (MLP) networks.

The set function f in Eq. 1 is invariant to input point permutations and can arbitrarily approximate any continuous set function [20]. Note that the response of h can be interpreted as the spatial encoding of a point (see [20] for details).

PointNet achieved impressive performance on a few benchmarks. However, it lacks the ability to capture local context at different scales. We will introduce a hierarchical feature learning framework in the next section to resolve the limitation.

3.2 Hierarchical Point Set Feature Learning

While PointNet uses a single max pooling operation to aggregate the whole point set, our new architecture builds a hierarchical grouping of points and progressively abstract larger and larger local regions along the hierarchy.

Our hierarchical structure is composed by a number of *set abstraction* levels (Fig. 2). At each level, a set of points is processed and abstracted to produce a new set with fewer elements. The set abstraction level is made of three key layers: *Sampling layer*, *Grouping layer* and *PointNet layer*. The *Sampling layer* selects a set of points from input points, which defines the centroids of local regions. *Grouping layer* then constructs local region sets by finding "neighboring" points around the centroids. *PointNet layer* uses a mini-PointNet to encode local region patterns into feature vectors.

A set abstraction level takes an $N \times (d + C)$ matrix as input that is from N points with d-dim coordinates and C-dim point feature. It outputs an $N' \times (d + C')$ matrix of N' subsampled points with d-dim coordinates and new C'-dim feature vectors summarizing local context. We introduce the layers of a set abstraction level in the following paragraphs.

Sampling layer. Given input points $\{x_1, x_2, ..., x_n\}$, we use iterative farthest point sampling (FPS) to choose a subset of points $\{x_{i_1}, x_{i_2}, ..., x_{i_m}\}$, such that x_{i_j} is the most distant point (in metric distance) from the set $\{x_{i_1}, x_{i_2}, ..., x_{i_{j-1}}\}$ with regard to the rest points. Compared with random sampling, it has better coverage of the entire point set given the same number of centroids. In contrast to CNNs that scan the vector space agnostic of data distribution, our sampling strategy generates receptive fields in a data dependent manner.

Grouping layer. The input to this layer is a point set of size $N \times (d + C)$ and the coordinates of a set of centroids of size $N' \times d$. The output are groups of point sets of size $N' \times K \times (d + C)$, where each group corresponds to a local region and K is the number of points in the neighborhood of centroid points. Note that K varies across groups but the succeeding *PointNet layer* is able to convert flexible number of points into a fixed length local region feature vector.

In convolutional neural networks, a local region of a pixel consists of pixels with array indices within certain Manhattan distance (kernel size) of the pixel. In a point set sampled from a metric space, the neighborhood of a point is defined by metric distance.

Ball query finds all points that are within a radius to the query point (an upper limit of K is set in implementation). An alternative range query is K nearest neighbor (kNN) search which finds a fixed

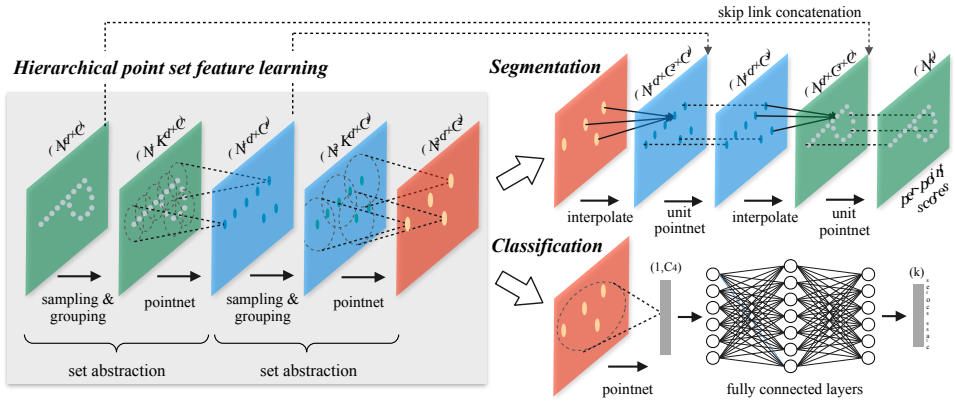


图2：我们的层次化特征学习架构及其在二维欧几里得空间中利用点进行集合分割与分类的应用示例。此处展示了单尺度点分组。有关密度自适应分组的详细信息，请参见图3。

其中 γ 和 h 通常是多层感知机（MLP）网络。

式(1)中的集合函数*f*对输入点的排列具有不变性，并且可以任意逼近任何连续集合函数[20]。注意，h的响应可以解释为点的空间编码（详见[20]）。

PointNet在少数基准测试上取得了令人瞩目的性能。然而，它缺乏在不同尺度上捕捉局部上下文的能力。我们将在下一节介绍一个层次化特征学习框架来解决这一局限。

3.2 分层点集特征学习

虽然PointNet使用单一最大池化操作来聚合整个点集，但我们的新架构构建了点的层次化分组，并沿层次结构逐步抽象出越来越大的局部区域。

我们的层次结构由若干*set abstraction*层级组成（图2）。在每个层级，一组点被处理并抽象化，以生成元素更少的新集合。集合抽象层级由三个关键层构成：*Sampling layer*、*Grouping layer*和*PointNet layer*。*Sampling layer*从输入点中选择一组点，这些点定义了局部区域的质心。*Grouping layer*随后通过查找质心周围的“邻近”点来构建局部区域集合。*PointNet layer*使用微型PointNet将局部区域模式编码为特征向量。

集合抽象层接收一个$N \times (d + C)$矩阵作为输入，该矩阵来自具有*d*维坐标和*C*维点特征的N个点。它输出一个N'个子采样点的$N' \times (d + C')$矩阵，包含*d*维坐标和新的C'维特征向量，用于汇总局部上下文。我们在以下段落中介绍集合抽象层的各层结构。

采样层。给定输入点集$\{x_1, x_2, ..., x_n\}$，我们使用迭代最远点采样（FPS）选取点的子集$\{x_{i_1}, x_{i_2}, ..., x_{i_m}\}$，使得$x_{i_j}$是相对于其余点而言距离集合$\{x_{i_1}, x_{i_2}, ..., x_{i_{j-1}}\}$最远的点（按度量距离）。与随机采样相比，在质心数量相同的情况下，该方法能更好地覆盖整个点集。与不关心数据分布而扫描向量空间的CNN不同，我们的采样策略以数据依赖的方式生成感受野。

分组层。该层的输入是一个大小为$N \times (d + C)$的点集，以及一组大小为$N' \times d$的质心坐标。输出是大小为$N' \times K \times (d + C)$的点集分组，其中每个分组对应一个局部区域，$K$是质心点邻域内的点数。请注意，$K$在不同分组之间变化，但后续的*PointNet layer*能够将灵活数量的点转换为固定长度的局部区域特征向量。

在卷积神经网络中，一个像素的局部区域由与该像素数组索引处于特定曼哈顿距离（核大小）内的像素组成。在从度量空间采样的点集中，一个点的邻域由度量距离定义。

球查询（ball query）会找出所有与查询点距离在半径范围内的点（实现中设置了上限 K）。另一种范围查询是 K 最近邻（kNN）搜索，它查找固定数量的

number of neighboring points. Compared with kNN, ball query's local neighborhood guarantees a fixed region scale thus making local region feature more generalizable across space, which is preferred for tasks requiring local pattern recognition (e.g. semantic point labeling).

PointNet layer. In this layer, the input are N' local regions of points with data size $N' \times K \times (d+C)$. Each local region in the output is abstracted by its centroid and local feature that encodes the centroid's neighborhood. Output data size is $N' \times (d + C')$.

The coordinates of points in a local region are firstly translated into a local frame relative to the centroid point: $x_i^{(j)} = x_i^{(j)} - \hat{x}^{(j)}$ for $i = 1, 2, ..., K$ and $j = 1, 2, ..., d$ where $\hat{x}$ is the coordinate of the centroid. We use PointNet [20] as described in Sec. 3.1 as the basic building block for local pattern learning. By using relative coordinates together with point features we can capture point-to-point relations in the local region.

3.3 Robust Feature Learning under Non-Uniform Sampling Density

As discussed earlier, it is common that a point set comes with non-uniform density in different areas. Such non-uniformity introduces a significant challenge for point set feature learning. Features learned in dense data may not generalize to sparsely sampled regions. Consequently, models trained for sparse point cloud may not recognize fine-grained local structures.

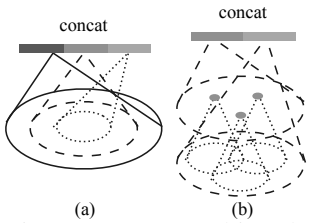


Figure 3: (a) Multi-scale grouping (MSG); (b) Multi-resolution grouping (MRG).

Ideally, we want to inspect as closely as possible into a point set to capture finest details in densely sampled regions. However, such close inspect is prohibited at low density areas because local patterns may be corrupted by the sampling deficiency. In this case, we should look for larger scale patterns in greater vicinity. To achieve this goal we propose density adaptive PointNet layers (Fig. 3) that learn to combine features from regions of different scales when the input sampling density changes. We call our hierarchical network with density adaptive PointNet layers as *PointNet++*.

Previously in Sec. 3.2, each abstraction level contains grouping and feature extraction of a single scale. In PointNet++, each abstraction level extracts multiple scales of local patterns and combine them intelligently according to local point densities. In terms of grouping local regions and combining features from different scales, we propose two types of density adaptive layers as listed below.

Multi-scale grouping (MSG). As shown in Fig. 3 (a), a simple but effective way to capture multi-scale patterns is to apply grouping layers with different scales followed by according PointNets to extract features of each scale. Features at different scales are concatenated to form a multi-scale feature.

We train the network to learn an optimized strategy to combine the multi-scale features. This is done by randomly dropping out input points with a randomized probability for each instance, which we call *random input dropout*. Specifically, for each training point set, we choose a dropout ratio θ uniformly sampled from $[0, p]$ where $p \leq 1$. For each point, we randomly drop a point with probability θ. In practice we set $p = 0.95$ to avoid generating empty point sets. In doing so we present the network with training sets of various sparsity (induced by θ) and varying uniformity (induced by randomness in dropout). During test, we keep all available points.

Multi-resolution grouping (MRG). The MSG approach above is computationally expensive since it runs local PointNet at large scale neighborhoods for every centroid point. In particular, since the number of centroid points is usually quite large at the lowest level, the time cost is significant.

Here we propose an alternative approach that avoids such expensive computation but still preserves the ability to adaptively aggregate information according to the distributional properties of points. In Fig. 3 (b), features of a region at some level L_i is a concatenation of two vectors. One vector (left in figure) is obtained by summarizing the features at each subregion from the lower level L_{i-1} using the set abstraction level. The other vector (right) is the feature that is obtained by directly processing all raw points in the local region using a single PointNet.

When the density of a local region is low, the first vector may be less reliable than the second vector, since the subregion in computing the first vector contains even sparser points and suffers more from sampling deficiency. In such a case, the second vector should be weighted higher. On the other hand,

邻域点的数量。与kNN相比，球查询的局部邻域保证了固定的区域尺度，从而使局部区域特征在空间上更具泛化能力，这对于需要局部模式识别（如语义点标注）的任务更为有利。

PointNet层。在该层中，输入为N'个点的局部区域，数据大小为$N' \times K \times (d+C)$。输出中的每个局部区域由其质心和编码质心邻域的局部特征进行抽象。输出数据大小为$N' \times (d + C')$。

局部区域中点的坐标首先被转换到相对于质心点的局部坐标系中：$x_i^{(j)} = x_i^{(j)} - \hat{x}^{(j)}$ 对应 $i = 1, 2, ..., K$，以及 $j = 1, 2, ..., d$，其中 $\hat{x}$ 是质心的坐标。我们采用第3.1节所述的PointNet [20]作为局部模式学习的基本构建模块。通过将相对坐标与点特征结合使用，我们能够捕捉局部区域中点与点之间的关系。

3.3 非均匀采样密度下的鲁棒特征学习

如前所述，点集在不同区域常具有非均匀密度。这种非均匀性给点集特征学习带来了重大挑战。在密集数据中学习到的特征可能无法泛化到稀疏采样区域。因此，针对稀疏点云训练的模型可能无法识别细粒度的局部结构。

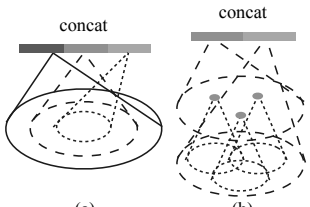


图3：(a) 多尺度分组（MSG）；(b) 多分辨率分组（MRG）。

理想情况下，我们希望尽可能深入地检查点集，以捕捉密集采样区域中最细微的细节。然而，在低密度区域，这种近距离检查是不可行的，因为局部模式可能因采样不足而受到破坏。在这种情况下，我们应在更大的邻域范围内寻找更大尺度的模式。为实现这一目标，我们提出了密度自适应PointNet层（图3），该层学习当输入采样密度变化时，结合不同尺度区域的特征。我们将这种带有密度自适应PointNet层的层次化网络称为*PointNet++*。

在之前的第3.2节中，每个抽象层级包含单一尺度的分组和特征提取。在PointNet++中，每个抽象层级提取多个尺度的局部模式，并根据局部点密度智能地组合它们。在分组局部区域和组合不同尺度特征方面，我们提出了两种类型的密度自适应层，如下所列。

多尺度分组（MSG）。如图3（a）所示，一种简单而有效的捕获多尺度模式的方法是应用不同尺度的分组层，随后使用相应的PointNets提取每个尺度的特征。不同尺度的特征被拼接起来形成多尺度特征。

我们训练网络学习一种优化策略，以组合多尺度特征。具体做法是，对每个实例以随机概率随机丢弃输入点，我们称之为*random input dropout*。具体而言，对于每个训练点集，我们从$[0, p]$中均匀采样一个丢弃比例θ，其中$p \leq 1$。对于每个点，我们以概率θ随机丢弃该点。实践中，我们设置$p = 0.95$以避免生成空点集。通过这种方式，我们向网络提供具有不同稀疏度（由θ引起）和不同均匀性（由丢弃过程中的随机性引起）的训练集。在测试时，我们保留所有可用点。

多分辨率分组（MRG）。上述MSG方法计算开销较大，因为它在每个质心点的大规模邻域上运行局部PointNet。特别是，由于最低层级上质心点的数量通常相当大，时间成本显著。

在此，我们提出一种替代方法，它避免了这种昂贵的计算，同时仍保留根据点的分布特性自适应聚合信息的能力。在图3(b)中，某一层级区域L_i的特征是两个向量的拼接。其中一个向量（图中左侧）是通过使用集合抽象层级对来自较低层级L_{i-1}的每个子区域的特征进行汇总而获得的。另一个向量（图中右侧）是使用单个PointNet直接处理局部区域内所有原始点而获得的特征。

当局部区域的密度较低时，第一个向量可能不如第二个向量可靠，因为计算第一个向量所涉及的子区域包含更稀疏的点，且受采样不足的影响更大。在这种情况下，第二个向量应被赋予更高的权重。另一方面，

when the density of a local region is high, the first vector provides information of finer details since it possesses the ability to inspect at higher resolutions recursively in lower levels.

Compared with MSG, this method is computationally more efficient since we avoids the feature extraction in large scale neighborhoods at lowest levels.

3.4 Point Feature Propagation for Set Segmentation

In set abstraction layer, the original point set is subsampled. However in set segmentation task such as semantic point labeling, we want to obtain point features for *all* the original points. One solution is to always sample all points as centroids in all set abstraction levels, which however results in high computation cost. Another way is to propagate features from subsampled points to the original points.

We adopt a hierarchical propagation strategy with distance based interpolation and across level skip links (as shown in Fig. 2). In a *feature propagation* level, we propagate point features from $N_l \times (d + C)$ points to N_{l-1} points where N_{l-1} and N_l (with $N_l \leq N_{l-1}$) are point set size of input and output of set abstraction level l. We achieve feature propagation by interpolating feature values f of N_l points at coordinates of the N_{l-1} points. Among the many choices for interpolation, we use inverse distance weighted average based on k nearest neighbors (as in Eq. 2, in default we use $p = 2$, $k = 3$). The interpolated features on N_{l-1} points are then concatenated with skip linked point features from the set abstraction level. Then the concatenated features are passed through a "unit pointnet", which is similar to one-by-one convolution in CNNs. A few shared fully connected and ReLU layers are applied to update each point's feature vector. The process is repeated until we have propagated features to the original set of points.

$$f^{(j)}(x) = \frac{\sum_{i=1}^{k} w_i(x) f_i^{(j)}}{\sum_{i=1}^{k} w_i(x)} \quad \text{where} \quad w_i(x) = \frac{1}{d(x, x_i)^p},\ j = 1, ..., C \tag{2}$$

4 Experiments

Datasets We evaluate on four datasets ranging from 2D objects (MNIST [11]), 3D objects (ModelNet40 [31] rigid object, SHREC15 [12] non-rigid object) to real 3D scenes (ScanNet [5]). Object classification is evaluated by accuracy. Semantic scene labeling is evaluated by average voxel classification accuracy following [5]. We list below the experiment setting for each dataset:

- MNIST: Images of handwritten digits with 60k training and 10k testing samples.
- ModelNet40: CAD models of 40 categories (mostly man-made). We use the official split with 9,843 shapes for training and 2,468 for testing.
- SHREC15: 1200 shapes from 50 categories. Each category contains 24 shapes which are mostly organic ones with various poses such as horses, cats, etc. We use five fold cross validation to acquire classification accuracy on this dataset.
- ScanNet: 1513 scanned and reconstructed indoor scenes. We follow the experiment setting in [5] and use 1201 scenes for training, 312 scenes for test.

4.1 Point Set Classification in Euclidean Metric Space

We evaluate our network on classifying point clouds sampled from both 2D (MNIST) and 3D (ModleNet40) Euclidean spaces. MNIST images are converted to 2D point clouds of digit pixel locations. 3D point clouds are sampled from mesh surfaces from ModelNet40 shapes. In default we use 512 points for MNIST and 1024 points for ModelNet40. In last row (ours normal) in Table 2, we use face normals as additional point features, where we also use more points ($N = 5000$) to further boost performance. All point sets are normalized to be zero mean and within a unit ball. We use a three-level hierarchical network with three fully connected layers [1]

Results. In Table 1 and Table 2, we compare our method with a representative set of previous state of the arts. Note that PointNet (vanilla) in Table 2 is the the version in [20] that does not use transformation networks, which is equivalent to our hierarchical net with only one level.

Firstly, our hierarchical learning architecture achieves significantly better performance than the non-hierarchical PointNet [20]. In MNIST, we see a relative 60.8% and 34.6% error rate reduction

[1] See supplementary for more details on network architecture and experiment preparation.

当局部区域的密度较高时，第一个向量提供更精细细节的信息，因为它具备在较低层级递归地以更高分辨率进行检查的能力。

与MSG相比，该方法在计算上更为高效，因为我们避免了在最低层级的大规模邻域中进行特征提取。

3.4 集合分割的点特征传播

在集合抽象层中，原始点集被下采样。然而，在诸如语义点标注等集合分割任务中，我们希望为*all*原始点获取点特征。一种解决方案是在所有集合抽象层级中始终将所有点采样为质心，但这会导致较高的计算成本。另一种方法是将特征从下采样点传播到原始点。

我们采用分层传播策略，结合基于距离的插值和跨层跳跃连接（如图2所示）。在*feature propagation*层中，我们将点特征从$N_l \times (d + C)$个点传播到N_{l-1}个点，其中N_{l-1}和N_l（$N_l \leq N_{l-1}$为集合抽象层l输入和输出的点集大小）实现特征传播。我们通过在N_{l-1}个点的坐标处插值N_l个点的特征值f来实现特征传播。在众多插值方法中，我们基于k个最近邻使用反距离加权平均（如公式2所示，默认情况下我们使用$p = 2$，$k = 3$）。N_{l-1}个点上的插值特征随后与来自集合抽象层的跳跃连接点特征进行拼接。然后，拼接后的特征通过一个“单元点网络”，该网络类似于CNN中的逐点卷积。应用若干共享的全连接层和ReLU层来更新每个点的特征向量。重复此过程，直到特征传播到原始点集。

$$f^{(j)}(x) = \frac{\sum_{i=1}^{k} w_i(x) f_i^{(j)}}{\sum_{i=1}^{k} w_i(x)} \quad \text{where} \quad w_i(x) = \frac{1}{d(x, x_i)^p},\ j = 1, ..., C \tag{2}$$

4 实验

数据集 我们在四个数据集上进行评估，涵盖从2D物体（MNIST [11]）、3D物体（ModelNet40 [31]刚性物体、SHREC15 [12]非刚性物体）到真实3D场景（ScanNet [5]）的各类数据。物体分类通过准确率进行评估。语义场景标注按照[5]中的方法，以平均体素分类准确率进行评估。以下列出每个数据集的实验设置：

- MNIST：包含6万训练样本和1万测试样本的手写数字图像。
- ModelNet40：40个类别（主要为人造物体）的CAD模型。我们使用官方划分，其中9,843个形状用于训练，2,468个用于测试。
- SHREC15：来自50个类别的1200个形状。每个类别包含24个形状，其中大部分是带有各种姿态的有机形状，如马、猫等。我们使用五折交叉验证来获取该数据集上的分类准确率。
- ScanNet：1513个扫描并重建的室内场景。我们遵循文献[5]中的实验设置，使用1201个场景进行训练，312个场景进行测试。

4.1 欧氏度量空间中的点集分类

我们在由2D（MNIST）和3D（ModelNet40）欧几里得空间采样的点云上评估我们的网络。MNIST图像被转换为数字像素位置的2D点云。3D点云从ModelNet40形状的网格表面采样。默认情况下，我们对MNIST使用512个点，对ModelNet40使用1024个点。在表2的最后一行（我们的法线）中，我们使用面法线作为额外的点特征，同时使用更多点（$N = 5000$）以进一步提升性能。所有点集均被归一化为零均值且位于单位球内。我们使用一个三级层次网络，包含三个全连接层[1]。

结果。在表1和表2中，我们将我们的方法与一组具有代表性的先前最优方法进行了比较。注意，表2中的PointNet（原始版本）是文献[20]中不使用变换网络的版本，这相当于我们仅包含一个层级的层次化网络。

首先，我们的分层学习架构比非分层PointNet[20]取得了显著更好的性能。在MNIST中，我们看到错误率相对降低了60.8%和34.6%。

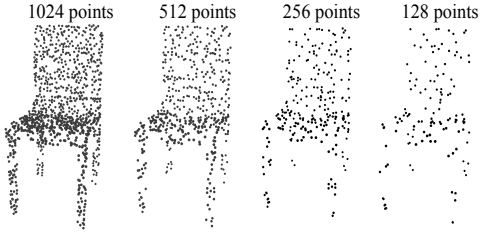
[1] See supplementary for more details on network architecture and experiment preparation.

Method	Error rate (%)
Multi-layer perceptron [24]	1.60
LeNet5 [11]	0.80
Network in Network [13]	**0.47**
PointNet (vanilla) [20]	1.30
PointNet [20]	0.78
Ours	0.51

Table 1: MNIST digit classification.

Method	Input	Accuracy (%)
Subvolume [21]	vox	89.2
MVCNN [26]	img	90.1
PointNet (vanilla) [20]	pc	87.2
PointNet [20]	pc	89.2
Ours	pc	90.7
Ours (with normal)	pc	**91.9**

Table 2: ModelNet40 shape classification.



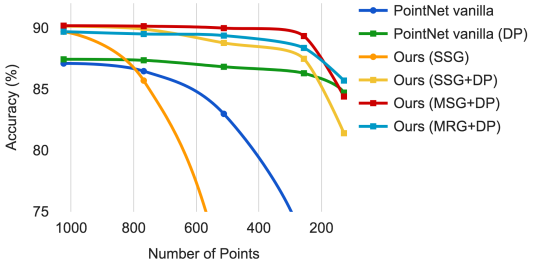


Figure 4: Left: Point cloud with random point dropout. Right: Curve showing advantage of our density adaptive strategy in dealing with non-uniform density. DP means random input dropout during training; otherwise training is on uniformly dense points. See Sec.3.3 for details.

from PointNet (vanilla) and PointNet to our method. In ModelNet40 classification, we also see that using same input data size (1024 points) and features (coordinates only), ours is remarkably stronger than PointNet. Secondly, we observe that point set based method can even achieve better or similar performance as mature image CNNs. In MNIST, our method (based on 2D point set) is achieving an accuracy close to the Network in Network CNN. In ModelNet40, ours with normal information significantly outperforms previous state-of-the-art method MVCNN [26].

Robustness to Sampling Density Variation. Sensor data directly captured from real world usually suffers from severe irregular sampling issues (Fig. 1). Our approach selects point neighborhood of multiple scales and learns to balance the descriptiveness and robustness by properly weighting them.

We randomly drop points (see Fig. 4 left) during test time to validate our network's robustness to non-uniform and sparse data. In Fig. 4 right, we see MSG+DP (multi-scale grouping with random input dropout during training) and MRG+DP (multi-resolution grouping with random input dropout during training) are very robust to sampling density variation. MSG+DP performance drops by less than 1% from 1024 to 256 test points. Moreover, it achieves the best performance on almost all sampling densities compared with alternatives. PointNet vanilla [20] is fairly robust under density variation due to its focus on global abstraction rather than fine details. However loss of details also makes it less powerful compared to our approach. SSG (ablated PointNet++ with single scale grouping in each level) fails to generalize to sparse sampling density while SSG+DP amends the problem by randomly dropping out points in training time.

4.2 Point Set Segmentation for Semantic Scene Labeling

To validate that our approach is suitable for large scale point cloud analysis, we also evaluate on semantic scene labeling task. The goal is to predict semantic object label for points in indoor scans. [5] provides a baseline using fully convolutional neural network on voxelized scans. They purely rely on scanning geometry instead of RGB information and report the accuracy on a per-voxel basis. To make a fair comparison, we remove RGB information in all our experiments and convert point cloud label prediction into voxel labeling following [5]. We also compare with [20]. The accuracy is reported on a per-voxel basis in Fig. 5 (blue bar).

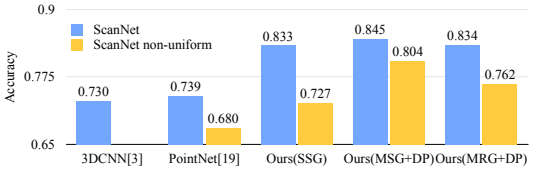


Figure 5: Scannet labeling accuracy.

Our approach outperforms all the baseline methods by a large margin. In comparison with [5], which learns on voxelized scans, we directly learn on point clouds to avoid additional quantization error,

Method	Error rate (%)
Multi-layer perceptron [24]	1.60
LeNet5 [11]	0.80
Network in Network [13]	**0.47**
PointNet (vanilla) [20]	1.30
PointNet [20]	0.78
Ours	0.51

Table 1: MNIST digit classification.

Method	Input	Accuracy (%)
Subvolume [21]	vox	89.2
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PointNet (vanilla) [20]	pc	87.2
PointNet [20]	pc	89.2
Ours	pc	90.7
Ours (with normal)	pc	**91.9**

Table 2: ModelNet40 shape classification.

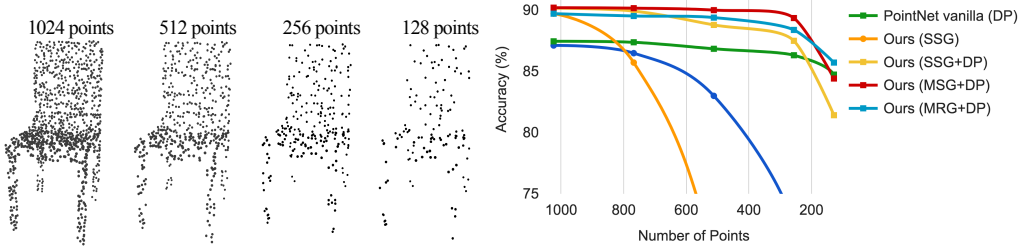


图4：左：具有随机点缺失的点云。右：曲线展示了我们的密度自适应策略在处理非均匀密度方面的优势。DP表示训练期间的随机输入缺失；否则训练基于均匀密集点。详见第3.3节。

从PointNet（原始版本）和PointNet到我们的方法。在ModelNet40分类中，我们还看到，在相同输入数据规模（1024个点）和特征（仅坐标）的情况下，我们的方法明显强于PointNet。其次，我们观察到基于点集的方法甚至可以达到与成熟的图像CNN相当或更好的性能。在MNIST中，我们的方法（基于2D点集）达到了接近Network in Network CNN的准确率。在ModelNet40中，我们的方法利用法线信息显著优于之前最先进的方法MVCNN [26]。

对采样密度变化的鲁棒性。从真实世界直接捕获的传感器数据通常存在严重的不规则采样问题（图1）。我们的方法选择多尺度的点邻域，并通过适当加权来学习平衡描述性与鲁棒性。

我们在测试时随机丢弃点（见图4左），以验证网络对非均匀和稀疏数据的鲁棒性。在图4右中，我们看到MSG+DP（训练时随机输入丢弃的多尺度分组）和MRG+DP（训练时随机输入丢弃的多分辨率分组）对采样密度变化非常鲁棒。MSG+DP在测试点从1024减少到256时，性能下降不到1%。此外，与替代方法相比，它在几乎所有采样密度下都取得了最佳性能。PointNet原始版本[20]在密度变化下相当鲁棒，因为它侧重于全局抽象而非精细细节。然而，细节的丢失也使其不如我们的方法强大。SSG（消融的PointNet++，每层使用单尺度分组）无法泛化到稀疏采样密度，而SSG+DP通过在训练时随机丢弃点来修正这一问题。

4.2 语义场景标注的点集分割

为了验证我们的方法适用于大规模点云分析，我们还在语义场景标注任务上进行了评估。其目标是为室内扫描中的点预测语义对象标签。[5] 提供了在体素化扫描上使用全卷积神经网络的基线方法。他们纯粹依赖扫描几何而非RGB信息，并报告了基于体素的准确率。为了进行公平比较，

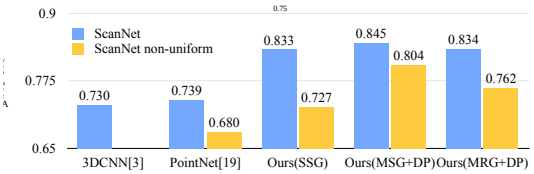


Figure 5: Scannet labeling accuracy.

我们在所有实验中移除RGB信息，并按照[5]将点云标签预测转换为体素标注。我们还与[20]进行了比较。图5（蓝色柱）中报告了基于逐体素的准确率。

我们的方法大幅优于所有基线方法。与在体素化扫描上学习的[5]相比，我们直接在点云上学习，以避免额外的量化误差，

and conduct data dependent sampling to allow more effective learning. Compared with [20], our approach introduces hierarchical feature learning and captures geometry features at different scales. This is very important for understanding scenes at multiple levels and labeling objects with various sizes. We visualize example scene labeling results in Fig. 6.

Robustness to Sampling Density Variation To test how our trained model performs on scans with non-uniform sampling density, we synthesize virtual scans of Scannet scenes similar to that in Fig. 1 and evaluate our network on this data. We refer readers to supplementary material for how we generate the virtual scans. We evaluate our framework in three settings (SSG, MSG+DP, MRG+DP) and compare with a baseline approach [20].

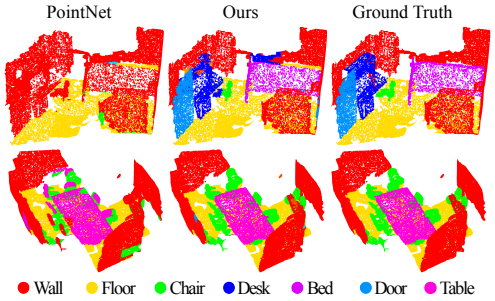


Figure 6: Scannet labeling results. [20] captures the overall layout of the room correctly but fails to discover the furniture. Our approach, in contrast, is much better at segmenting objects besides the room layout.

Performance comparison is shown in Fig. 5 (yellow bar). We see that SSG performance greatly falls due to the sampling density shift from uniform point cloud to virtually scanned scenes. MRG network, on the other hand, is more robust to the sampling density shift since it is able to automatically switch to features depicting coarser granularity when the sampling is sparse. Even though there is a domain gap between training data (uniform points with random dropout) and scanned data with non-uniform density, our MSG network is only slightly affected and achieves the best accuracy among methods in comparison. These prove the effectiveness of our density adaptive layer design.

4.3 Point Set Classification in Non-Euclidean Metric Space

In this section, we show generalizability of our approach to non-Euclidean space. In non-rigid shape classification (Fig. 7), a good classifier should be able to classify (a) and (c) in Fig. 7 correctly as the same category even given their difference in pose, which requires knowledge of intrinsic structure. Shapes in SHREC15 are 2D surfaces embedded in 3D space. Geodesic distances along the surfaces naturally induce a metric space. We show through experiments that adopting PointNet++ in this metric space is an effective way to capture intrinsic structure of the underlying point set.

For each shape in [12], we firstly construct the metric space induced by pairwise geodesic distances. We follow [23] to obtain an embedding metric that mimics geodesic distance. Next we extract intrinsic point features in this metric space including WKS [1], HKS [27] and multi-scale Gaussian curvature [16]. We use these features as input and then sample and group points according to the underlying metric space. In this way, our network learns to capture multi-scale intrinsic structure that is not influenced by the specific pose of a shape. Alternative design choices include using XYZ coordinates as points feature or use Euclidean space $\mathbb{R}^3$ as the underlying metric space. We show below these are not optimal choices.


(a) Horse (b) Cat (c) Horse
Figure 7: An example of non-rigid shape classification.

Results. We compare our methods with previous state-of-the-art method [14] in Table 3. [14] extracts geodesic moments as shape features and use a stacked sparse autoencoder to digest these features to predict shape category. Our approach using non-Euclidean metric space and intrinsic features achieves the best performance in all settings and outperforms [14] by a large margin.

Comparing the first and second setting of our approach, we see intrinsic features are very important for non-rigid shape classification. XYZ feature fails to reveal intrinsic structures and is greatly influenced by pose variation. Comparing the second and third setting of our approach, we see using geodesic neighborhood is beneficial compared with Euclidean neighborhood. Euclidean neighborhood might include points far away on surfaces and this neighborhood could change dramatically when shape affords non-rigid deformation. This introduces difficulty for effective weight sharing since the local structure could become combinatorially complicated. Geodesic neighborhood on surfaces, on the other hand, gets rid of this issue and improves the learning effectiveness.

并进行数据依赖采样，以实现更有效的学习。与[20]相比，我们的方法引入了层次化特征学习，并在不同尺度上捕捉几何特征。这对于多层次理解场景以及标注不同大小的物体非常重要。我们在图6中可视化了示例场景标注结果。

对采样密度变化的鲁棒性 为测试我们训练好的模型在非均匀采样密度扫描上的表现，我们合成了与图1类似的Scannet场景虚拟扫描，并在此数据上评估我们的网络。关于虚拟扫描的生成方式，我们请读者参阅补充材料。我们在三种设置（SSG、MSG+DP、MRG+DP）下评估我们的框架，并与基线方法[20]进行比较。

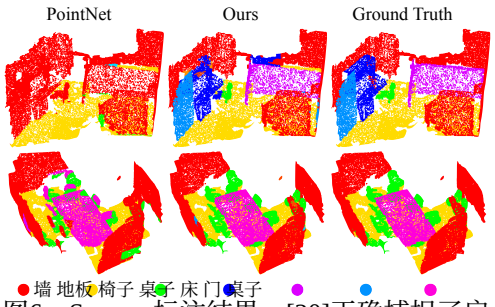


图6：Scannet标注结果。[20]正确捕捉了房间的整体布局，但未能发现家具。相比之下，我们的方法在分割房间布局之外的物体方面要好得多。

性能对比见图5（黄色柱）。我们看到，由于采样密度从均匀点云转变为虚拟扫描场景，SSG的性能大幅下降。相比之下，MRG网络对采样密度变化更为鲁棒，因为它能够在采样稀疏时自动切换到描述更粗粒度的特征。尽管存在域

训练数据（均匀点且随机丢失）与扫描数据（密度非均匀）之间的差距，我们的MSG网络仅受到轻微影响，并在对比方法中取得了最佳精度。这些证明了我们的密度自适应层设计的有效性。

4.3 非欧几里得度量空间中的点集分类 ce

在本节中，我们展示了我们的方法在非欧几里得空间中的泛化能力。在非刚性形状分类中（图7），一个好的分类器应能将图7中的(*a*)和(*c*)正确分类为同一类别，即使它们在姿态上存在差异，这需要对其内在结构有所了解。SHREC15中的形状是嵌入在三维空间中的二维曲面。沿曲面的测地距离自然诱导出一个度量空间。我们通过实验表明，在此度量空间中采用PointNet++是捕捉底层点集内在结构的有效方式。

对于[12]中的每个形状，我们首先构造由两两测地距离诱导的度量空间。我们遵循[23]获得一种模拟测地距离的嵌入度量。接着，我们在此度量空间中提取内在点特征，包括WKS [1]、HKS [27]和多尺度高斯曲率[16]。我们将这些特征作为输入，然后根据底层度量空间对点进行采样和分组。通过这种方式，我们的网络学会捕获不受形状特定姿态影响的多尺度内在结构。其他可选设计方案包括使用XYZ坐标作为点特征，或使用欧几里得空间$\mathbb{R}^3$作为底层度量空间。我们在下文表明这些并非最优选择。


(a) Horse (b) Cat (c) Horse
图7：非刚性形状分类的一个示例。

结果。我们在表3中将我们的方法与先前的最先进方法[14]进行了比较。[14]提取测地矩作为形状特征，并使用堆叠稀疏自编码器来消化这些特征以预测形状类别。我们的方法采用非欧几里得度量空间和内在特征，在所有设置中均取得了最佳性能，并以较大优势超越了[14]。

比较我们方法的第一种和第二种设置，可以看出内在特征对于非刚性形状分类非常重要。XYZ特征无法揭示内在结构，且受姿态变化影响很大。比较第二种和第三种设置，可以看出使用测地邻域相比欧几里得邻域更有益。欧几里得邻域可能包含表面上距离较远的点，并且当形状发生非刚性变形时，该邻域可能发生剧烈变化。这给有效的权重共享带来了困难，因为局部结构可能变得组合上非常复杂。相比之下，表面上的测地邻域则避免了这一问题，并提高了学习效果。

	Metric space	Input feature	Accuracy (%)
DeepGM [14]	-	Intrinsic features	93.03
Ours	Euclidean	XYZ	60.18
	Euclidean	Intrinsic features	94.49
	Non-Euclidean	Intrinsic features	**96.09**

Table 3: SHREC15 Non-rigid shape classification.

4.4 Feature Visualization.

In Fig. 8 we visualize what has been learned by the first level kernels of our hierarchical network. We created a voxel grid in space and aggregate local point sets that activate certain neurons the most in grid cells (highest 100 examples are used). Grid cells with high votes are kept and converted back to 3D point clouds, which represents the pattern that neuron recognizes. Since the model is trained on ModelNet40 which is mostly consisted of furniture, we see structures of planes, double planes, lines, corners etc. in the visualization.

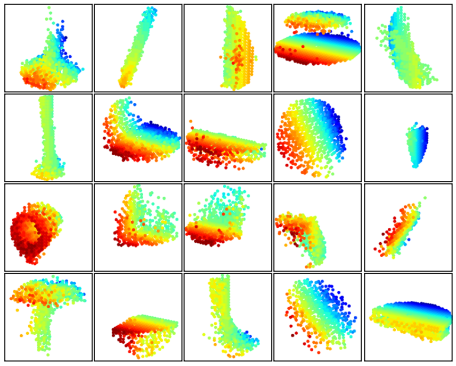


Figure 8: 3D point cloud patterns learned from the first layer kernels. The model is trained for ModelNet40 shape classification (20 out of the 128 kernels are randomly selected). Color indicates point depth (red is near, blue is far).

5 Related Work

The idea of hierarchical feature learning has been very successful. Among all the learning models, convolutional neural network [10, 25, 8] is one of the most prominent ones. However, convolution does not apply to unordered point sets with distance metrics, which is the focus of our work.

A few very recent works [20, 28] have studied how to apply deep learning to unordered sets. They ignore the underlying distance metric even if the point set does possess one. As a result, they are unable to capture local context of points and are sensitive to global set translation and normalization. In this work, we target at points sampled from a metric space and tackle these issues by explicitly considering the underlying distance metric in our design.

Point sampled from a metric space are usually noisy and with non-uniform sampling density. This affects effective point feature extraction and causes difficulty for learning. One of the key issue is to select proper scale for point feature design. Previously several approaches have been developed regarding this [19, 17, 2, 6, 7, 30] either in geometry processing community or photogrammetry and remote sensing community. In contrast to all these works, our approach learns to extract point features and balance multiple feature scales in an end-to-end fashion.

In 3D metric space, other than point set, there are several popular representations for deep learning, including volumetric grids [21, 22, 29], and geometric graphs [3, 15, 33]. However, in none of these works, the problem of non-uniform sampling density has been explicitly considered.

6 Conclusion

In this work, we propose PointNet++, a powerful neural network architecture for processing point sets sampled in a metric space. PointNet++ recursively functions on a nested partitioning of the input point set, and is effective in learning hierarchical features with respect to the distance metric. To handle the non uniform point sampling issue, we propose two novel set abstraction layers that intelligently aggregate multi-scale information according to local point densities. These contributions enable us to achieve state-of-the-art performance on challenging benchmarks of 3D point clouds.

In the future, it's worthwhile thinking how to accelerate inference speed of our proposed network especially for MSG and MRG layers by sharing more computation in each local regions. It's also interesting to find applications in higher dimensional metric spaces where CNN based method would be computationally unfeasible while our method can scale well.

	Metric space	Input feature	Accuracy (%)
DeepGM [14]	-	Intrinsic features	93.03
Ours	Euclidean	XYZ	60.18
	Euclidean	Intrinsic features	94.49
	Non-Euclidean	Intrinsic features	**96.09**

表3：SHREC15非刚性形状分类。

4.4 特征可视化。

在图8中，我们可视化了分层网络第一层核所学习到的内容。我们在空间中创建了一个体素网格，并聚合在网格单元中最能激活某些神经元的局部点集（使用得分最高的100个示例）。具有高投票数的网格单元被保留并转换回3D点云，这代表了该神经元识别的模式。由于模型是在ModelNet40上训练的，该数据集主要由家具组成，因此我们在可视化中看到了平面、双平面、线条、角落等结构。

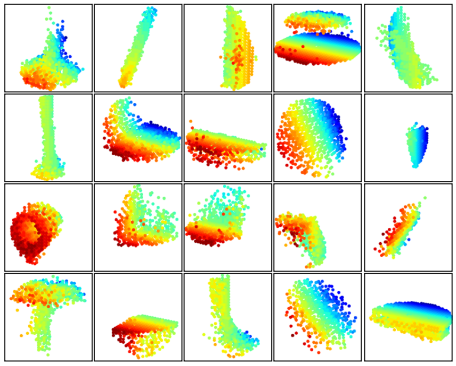


图8：从第一层核中学习到的三维点云模式。该模型针对ModelNet40形状分类任务进行训练（从128个核中随机选取20个）。颜色表示点的深度（红色为近，蓝色为远）。

5 相关工作

分层特征学习的思想非常成功。在所有学习模型中，卷积神经网络[10, 25, 8]是最突出的模型之一。然而，卷积并不适用于具有距离度量的无序点集，而这正是我们工作的重点。

一些非常近期的工作[20, 28]研究了如何
将深度学习应用于无序集合。即使点集确实具有距离度量，它们也会忽略底层距离度量。因此，它们无法捕捉点的局部上下文，并且对全局集合平移和归一化敏感。在这项工作中，我们针对从度量空间采样的点，并通过在设计中有意考虑底层距离度量来解决这些问题。

从度量空间中采样的点通常带有噪声，且采样密度不均匀。这影响了有效的点特征提取，并给学习带来困难。关键问题之一是为点特征设计选择合适的尺度。此前，已有若干方法针对这一问题展开研究[19, 17, 2, 6, 7, 30]，这些工作或来自几何处理领域，或来自摄影测量与遥感领域。与所有这些工作不同，我们的方法以端到端的方式学习提取点特征并平衡多个特征尺度。

在三维度量空间中，除点集外，还有几种流行的深度学习表示方法，包括体素网格[21, 22, 29]和几何图[3, 15, 33]。然而，在这些工作中，均未明确考虑非均匀采样密度的问题。

6 结论

在本工作中，我们提出了PointNet++，一种用于处理度量空间中采样点集的强大神经网络架构。PointNet++在输入点集的嵌套划分上递归运行，并能有效学习与距离度量相关的层次化特征。为解决非均匀点采样问题，我们提出了两种新颖的集合抽象层，可根据局部点密度智能聚合多尺度信息。这些贡献使我们能够在三维点云的挑战性基准上实现最先进的性能。

未来，值得思考如何加速我们提出的网络的推理速度，尤其是MSG和MRG层，通过在各个局部区域共享更多计算来实现。此外，探索在更高维度量空间中的应用也很有趣，在这些空间中，基于CNN的方法在计算上不可行，而我们的方法能够很好地扩展。

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Supplementary

A Overview

This supplementary material provides more details on experiments in the main paper and includes more experiments to validate and analyze our proposed method.

In Sec B we provide specific network architectures used for experiments in the main paper and also describe details in data preparation and training. In Sec C we show more experimental results including benchmark performance on part segmentation and analysis on neighborhood query, sensitivity to sampling randomness and time space complexity.

B Details in Experiments

Architecture protocol. We use following notations to describe our network architecture.

SA(K,r,[l_1, ..., l_d]) is a set abstraction (SA) level with K local regions of ball radius r using PointNet of d fully connected layers with width l_i ($i = 1, ..., d$). SA([l_1, ...l_d]) is a global set abstraction level that converts set to a single vector. In multi-scale setting (as in MSG), we use SA(K, [$r^{(1)}$, ..., $r^{(m)}$], [[$l_1^{(1)}$, ..., $l_d^{(1)}$],...,[$l_1^{(m)}$, ..., $l_d^{(m)}$]]) to represent MSG with m scales.

FC(l,dp) represents a fully connected layer with width l and dropout ratio dp. FP(l_1, ..., l_d) is a feature propagation (FP) level with d fully connected layers. It is used for updating features concatenated from interpolation and skip link. All fully connected layers are followed by batch normalization and ReLU except for the last score prediction layer.

B.1 Network Architectures

For all classification experiments we use the following architecture (Ours SSG) with different K (number of categories):

$$SA(512, 0.2, [64, 64, 128]) \rightarrow SA(128, 0.4, [128, 128, 256]) \rightarrow SA([256, 512, 1024]) \rightarrow FC(512, 0.5) \rightarrow FC(256, 0.5) \rightarrow FC(K)$$

The multi-scale grouping (MSG) network (PointNet++) architecture is as follows:

$$SA(512, [0.1, 0.2, 0.4], [[32, 32, 64], [64, 64, 128], [64, 96, 128]]) \rightarrow SA(128, [0.2, 0.4, 0.8], [[64, 64, 128], [128, 128, 256], [128, 128, 256]]) \rightarrow SA([256, 512, 1024]) \rightarrow FC(512, 0.5) \rightarrow FC(256, 0.5) \rightarrow FC(K)$$

The cross level multi-resolution grouping (MRG) network's architecture uses three branches:

Branch 1: $SA(512, 0.2, [64, 64, 128]) \rightarrow SA(64, 0.4, [128, 128, 256])$

Branch 2: $SA(512, 0.4, [64, 128, 256])$ using $r = 0.4$ regions of original points

Branch 3: $SA(64, 128, 256, 512)$ using all original points.

Branch 4: $SA(256, 512, 1024)$.

Branch 1 and branch 2 are concatenated and fed to branch 4. Output of branch 3 and branch4 are then concatenated and fed to $FC(512, 0.5) \rightarrow FC(256, 0.5) \rightarrow FC(K)$ for classification.

Network for semantic scene labeling (last two fully connected layers in FP are followed by dropout layers with drop ratio 0.5):

$$SA(1024, 0.1, [32, 32, 64]) \rightarrow SA(256, 0.2, [64, 64, 128]) \rightarrow SA(64, 0.4, [128, 128, 256]) \rightarrow SA(16, 0.8, [256, 256, 512]) \rightarrow FP(256, 256) \rightarrow FP(256, 256) \rightarrow FP(256, 128) \rightarrow FP(128, 128, 128, 128, K)$$

Network for semantic and part segmentation (last two fully connected layers in FP are followed by dropout layers with drop ratio 0.5):

$$SA(512, 0.2, [64, 64, 128]) \rightarrow SA(128, 0.4, [128, 128, 256]) \rightarrow SA([256, 512, 1024]) \rightarrow FP(256, 256) \rightarrow FP(256, 128) \rightarrow FP(128, 128, 128, 128, K)$$

补充材料

A 概述

本补充材料提供了主论文中实验的更多细节，并包含更多实验来验证和分析我们提出的方法。

在B节中，我们提供了主论文实验中使用的具体网络架构，并描述了数据准备和训练的细节。在C节中，我们展示了更多实验结果，包括部分分割的基准性能、邻域查询分析、对采样随机性的敏感性以及时空复杂度。

B 实验中的细节

架构协议。我们使用以下符号来描述我们的网络架构。

SA(K,r,[$l_1, ..., l_d$])是一个集合抽象（SA）层级，包含K个半径为r的局部区域，使用具有d个全连接层、宽度为l_i ($i = 1, ..., d$)的PointNet。SA([$l_1, ...l_d$])是一个全局集合抽象层级，将集合转换为单个向量。在多尺度设置（如MSG）中，我们使用SA(K, [$r^{(1)}, ..., r^{(m)}$], [[$l_1^{(1)}, ..., l_d^{(1)}$],...,[$l_1^{(m)}, ..., l_d^{(m)}$]])来表示具有m个尺度的MSG。

FC(l,dp)表示宽度为l、丢弃率为dp的全连接层。FP($l_1, ..., l_d$)是一个具有d个全连接层的特征传播（FP）层，用于更新由插值和跳跃连接拼接得到的特征。除最后的分数预测层外，所有全连接层后均接有批归一化和ReLU。

B.1 网络架构

对于所有分类实验，我们使用以下架构（我们的SSG），其中包含不同K (类别数量):

$$SA(512, 0.2, [64, 64, 128]) \rightarrow SA(128, 0.4, [128, 128, 256]) \rightarrow SA([256, 512, 1024]) \rightarrow FC(512, 0.5) \rightarrow FC(256, 0.5) \rightarrow FC(K)$$

多尺度分组（MSG）网络（PointNet++）架构如下:

SA(512, [0.1, 0.2, 0.4], [[32, 32, 64], [64, 64, 128], [64, 96, 128]]) $\rightarrow$ SA(128, [0.2, 0.4, 0.8], [[64, 64, 128], [128, 128, 256], [128, 128, 256]]) $\rightarrow$ SA([256, 512, 1024]) $\rightarrow$ FC(512, 0.5) $\rightarrow$ FC(256, 0.5) $\rightarrow$ $FC(K)$

跨层级多分辨率分组（MRG）网络的架构使用三个分支:

Branch 1: $SA(512, 0.2, [64, 64, 128]) \rightarrow SA(64, 0.4, [128, 128, 256])$

分支2：SA(512, 0.4, [64, 128, 256]) 使用原始点的$r = 0.4$个区域

分支3：SA(64, 128, 256, 512)，使用所有原始点。分

支4：SA(256, 512, 1024)。

分支1和分支2被拼接后输入分支4。分支3和分支4的输出随后被拼接并输入到FC(512, 0.5) $\rightarrow$ FC(256, 0.5) $\rightarrow$ $FC(K)$进行分类。

用于语义场景标注的网络（FP中最后两个全连接层后接丢弃率为0.5的丢弃层）：

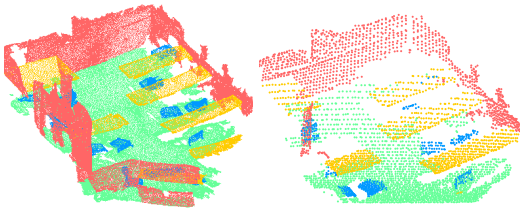
$$SA(1024, 0.1, [32, 32, 64]) \rightarrow SA(256, 0.2, [64, 64, 128]) \rightarrow SA(64, 0.4, [128, 128, 256]) \rightarrow SA(16, 0.8, [256, 256, 512]) \rightarrow FP(256, 256) \rightarrow FP(256, 256) \rightarrow FP(256, 128) \rightarrow FP(128, 128, 128, 128, K)$$

用于语义和部件分割的网络（FP中最后两个全连接层后接丢弃率为0.5的丢弃层）：

SA(512, 0.2, [64, 64, 128]) $\rightarrow$ SA(128, 0.4, [128, 128, 256]) $\rightarrow$ SA([256, 512, 1024]) $\rightarrow$ FP(256, 256) $\rightarrow$ FP(256, 128) $\rightarrow$ FP(128, 128, 128, 128, K)

B.2 Virtual Scan Generation

In this section, we describe how we generate labeled virtual scan with non-uniform sampling density from ScanNet scenes. For each scene in ScanNet, we set camera location $1.5m$ above the centroid of the floor plane and rotate the camera orientation in the horizontal plane evenly in 8 directions. In each direction, we use a image plane with size $100px$ by $75px$ and cast rays from camera through each pixel to the scene. This gives a way to select visible points in the scene. We could then generate 8 virtual scans for each test scene similar and an example is shown in Fig. 9. Notice point samples are denser in regions closer to the camera.



(a) ScanNet labeled scene (b) ScanNet non-uniform

Figure 9: Virtual scan generated from ScanNet

B.3 MNIST and ModelNet40 Experiment Details

For MNIST images, we firstly normalize all pixel intensities to range $[0, 1]$ and then select all pixels with intensities larger than 0.5 as valid digit pixels. Then we convert digit pixels in an image into a 2D point cloud with coordinates within $[-1, 1]$, where the image center is the origin point. Augmented points are created to add the point set up to a fixed cardinality (512 in our case). We jitter the initial point cloud (with random translation of Gaussian distribution $\mathcal{N}(0, 0.01)$ and clipped to 0.03) to generate the augmented points. For ModelNet40, we uniformly sample N points from CAD models surfaces based on face area.

For all experiments, we use Adam [9] optimizer with learning rate 0.001 for training. For data augmentation, we randomly scale object, perturb the object location as well as point sample locations. We also follow [21] to randomly rotate objects for ModelNet40 data augmentation. We use TensorFlow and GTX 1080, Titan X for training. All layers are implemented in CUDA to run GPU. It takes around 20 hours to train our model to convergence.

B.4 ScanNet Experiment Details

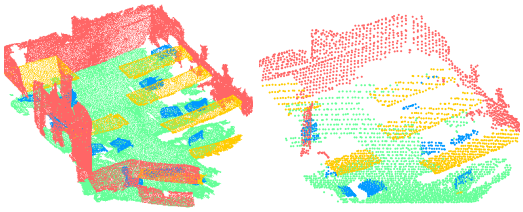
To generate training data from ScanNet scenes, we sample 1.5m by 1.5m by 3m cubes from the initial scene and then keep the cubes where $\geq 2\%$ of the voxels are occupied and $\geq 70\%$ of the surface voxels have valid annotations (this is the same set up in [5]). We sample such training cubes on the fly and random rotate it along the up-right axis. Augmented points are added to the point set to make a fixed cardinality (8192 in our case). During test time, we similarly split the test scene into smaller cubes and get label prediction for every point in the cubes first, then merge label prediction in all the cubes from a same scene. If a point get different labels from different cubes, we will just conduct a majority voting to get the final point label prediction.

B.5 SHREC15 Experiment Details

We randomly sample 1024 points on each shape both for training and testing. To generate the input intrinsic features, we to extract 100 dimensional WKS, HKS and multiscale Gaussian curvature respectively, leading to a 300 dimensional feature vector for each point. Then we conduct PCA to reduce the feature dimension to 64. We use a 8 dimensional embedding following [23] to mimic the geodesic distance, which is used to describe our non-Euclidean metric space while choosing the point neighborhood.

B.2 虚拟扫描生成

在本节中，我们描述如何从ScanNet场景生成具有非均匀采样密度的带标注虚拟扫描。对于ScanNet中的每个场景，我们将相机位置设置在楼层平面质心上方1.5m处，并在水平面上沿8个方向均匀旋转相机朝向。在每个方向上，我们使用大小为100px×75px的图像平面，并从相机出发穿过每个像素向场景投射光线。这样可以选择场景中的可见点。随后，我们可以为每个测试场景生成8个类似的虚拟扫描，图9展示了一个示例。请注意，靠近相机的区域点采样更密集。



(a) ScanNet labeled scene　　(b) ScanNet non-uniform

图9：由ScanNet生成的虚拟扫描

B.3 MNIST与ModelNet40实验细节

对于MNIST图像，我们首先将所有像素强度归一化到范围[,0, 1]，然后选择强度大于0.5的所有像素作为有效数字像素。接着，我们将图像中的数字像素转换为二维点云，其坐标位于[−-1, ,1]范围内，其中图像中心为原点。通过增广点将点集补充至固定基数（本例中为512）。我们对初始点云进行抖动处理（采用高斯分布$\mathcal{N}$(0, ,0, .01的随机平移，并裁剪至0.03）以生成增广点。对于ModelNet40，我们基于面面积从CAD模型表面均匀采样N个点。

所有实验中，我们使用Adam [9]优化器，学习率设为0.001进行训练。对于数据增强，我们随机缩放物体、扰动物体位置以及点采样位置。我们还遵循[21]中的方法，对ModelNet40数据增强随机旋转物体。我们使用Tensor-Flow和GTX 1080、Titan X进行训练。所有层均用CUDA实现以在GPU上运行。训练模型至收敛大约需要20小时。

B.4 ScanNet实验细节

为了从ScanNet场景生成训练数据，我们从初始场景中采样1.5米×1.5米×3米的立方体，然后保留满足≥ 2%体素被占用且≥ 70%表面体素具有有效标注的立方体（这与[5]中的设置相同）。我们动态采样此类训练立方体，并沿竖直轴随机旋转。向点集中添加增强点以保持固定基数（本例中为8192）。在测试时，我们同样将测试场景分割为较小的立方体，首先获取每个立方体中每个点的标签预测，然后合并来自同一场景所有立方体的标签预测。如果某个点从不同立方体获得不同标签，我们将进行多数投票以获得最终的点标签预测。

B.5 SHREC15实验细节

我们在每个形状上随机采样1024个点，用于训练和测试。为生成输入的内在特征，我们分别提取100维的WKS、HKS和多尺度高斯曲率，从而为每个点生成300维的特征向量。随后，我们进行PCA将特征维度降至64。我们采用8维嵌入，参照[23]来模拟测地距离，该距离用于描述我们的非欧几里得度量空间，同时用于选择点的邻域。

C More Experiments

In this section we provide more experiment results to validate and analyze our proposed network architecture.

C.1 Semantic Part Segmentation

Following the setting in [32], we evaluate our approach on part segmentation task assuming category label for each shape is already known. Taken shapes represented by point clouds as input, the task is to predict a part label for each point. The dataset contains 16,881 shapes from 16 classes, annotated with 50 parts in total. We use the official train test split following [4].

We equip each point with its normal direction to better depict the underlying shape. This way we could get rid of hand-crafted geometric features as is used in [32, 33]. We compare our framework with traditional learning based techniques [32], as well as state-of-the-art deep learning approaches [20, 33] in Table 4. Point intersection over union (IoU) is used as the evaluation metric, averaged across all part classes. Cross-entropy loss is minimized during training. On average, our approach achieves the best performance. In comparison with [20], our approach performs better on most of the categories, which proves the importance of hierarchical feature learning for detailed semantic understanding. Notice our approach could be viewed as implicitly building proximity graphs at different scales and operating on these graphs, thus is related to graph CNN approaches such as [33]. Thanks to the flexibility of our multi-scale neighborhood selection as well as the power of set operation units, we could achieve better performance compared with [33]. Notice our set operation unit is much simpler compared with graph convolution kernels, and we do not need to conduct expensive eigen decomposition as opposed to [33]. These make our approach more suitable for large scale point cloud analysis.

	mean	aero	bag	cap	car	chair	ear phone	guitar	knife	lamp	laptop	motor	mug	pistol	rocket	skate board	table
Yi [32]	81.4	81.0	78.4	77.7	75.7	87.6	61.9	92.0	85.4	82.5	95.7	70.6	91.9	85.9	53.1	69.8	75.3
PN [20]	83.7	83.4	78.7	82.5	74.9	89.6	73.0	91.5	85.9	80.8	95.3	65.2	93.0	81.2	57.9	72.8	80.6
SSCNN [33]	84.7	81.6	81.7	81.9	75.2	90.2	74.9	93.0	86.1	84.7	95.6	66.7	92.7	81.6	60.6	82.9	82.1
Ours	85.1	82.4	79.0	87.7	77.3	90.8	71.8	91.0	85.9	83.7	95.3	71.6	94.1	81.3	58.7	76.4	82.6

Table 4: Segmentation results on ShapeNet part dataset.

C.2 Neighborhood Query: kNN v.s. Ball Query.

Here we compare two options to select a local neighborhood. We used radius based ball query in our main paper. Here we also experiment with kNN based neighborhood search and also play with different search radius and k. In this experiment all training and testing are on ModelNet40 shapes with uniform sampling density. 1024 points are used. As seen in Table 5, radius based ball query is slightly better than kNN based method. However, we speculate in very non-uniform point set, kNN based query will results in worse generalization ability. Also we observe that a slightly large radius is helpful for performance probably because it captures richer local patterns.

kNN (k=16)	kNN (k=64)	radius (r=0.1)	radius (r=0.2)
89.3	90.3	89.1	90.7

Table 5: Effects of neighborhood choices. Evaluation metric is classification accuracy (%) on ModelNet 40 test set.

C.3 Effect of Randomness in Farthest Point Sampling.

For the *Sampling layer* in our set abstraction level, we use farthest point sampling (FPS) for point set sub sampling. However FPS algorithm is random and the subsampling depends on which point is selected first. Here we evaluate the sensitivity of our model to this randomness. In Table 6, we test our model trained on ModelNet40 for feature stability and classification stability.

C 更多实验

在本节中，我们提供更多实验结果来验证和分析我们提出的网络架构。

C.1 语义部分分割

遵循文献[32]的设置，我们在部件分割任务上评估我们的方法，并假设每个形状的类别标签已知。以点云表示的形状作为输入，任务是为每个点预测一个部件标签。该数据集包含来自16个类别的16,881个形状，共标注了50个部件。我们采用文献[4]中的官方训练测试划分。

我们为每个点配备其法线方向，以更好地描绘底层形状。这样，我们就能摆脱[32, 33]中使用的手工几何特征。我们在表4中将我们的框架与传统的基于学习的技术[32]以及最先进的深度学习方法[20, 33]进行了比较。采用点交并比（IoU）作为评估指标，并在所有部件类别上取平均。训练过程中最小化交叉熵损失。平均而言，我们的方法取得了最佳性能。与[20]相比，我们的方法在大多数类别上表现更好，这证明了层次化特征学习对于细粒度语义理解的重要性。请注意，我们的方法可以被视为在不同尺度上隐式构建邻近图并在此类图上操作，因此与图卷积网络方法（如[33]）相关。得益于我们多尺度邻域选择的灵活性以及集合操作单元的强大能力，我们能够取得比[33]更好的性能。还需注意，我们的集合操作单元比图卷积核简单得多，并且与[33]不同，我们无需进行昂贵的特征分解。这些特点使我们的方法更适合大规模点云分析。

	mean	aero	bag	cap	car	chair	ear phone	guitar	knife	lamp	laptop	motor	mug	pistol	rocket	skate board	table
Yi [32]	81.4	81.0	78.4	77.7	75.7	87.6	61.9	92.0	85.4	82.5	95.7	70.6	91.9	85.9	53.1	69.8	75.3
PN [20]	83.7	83.4	78.7	82.5	74.9	89.6	73.0	91.5	85.9	80.8	95.3	65.2	93.0	81.2	57.9	72.8	80.6
SSCNN [33]	84.7	81.6	81.7	81.9	75.2	90.2	74.9	93.0	86.1	84.7	95.6	66.7	92.7	81.6	60.6	82.9	82.1
Ours	85.1	82.4	79.0	87.7	77.3	90.8	71.8	91.0	85.9	83.7	95.3	71.6	94.1	81.3	58.7	76.4	82.6

表4：ShapeNet部件数据集上的分割结果。

C.2 邻域查询：kNN 与球查询。

在此我们比较两种选择局部邻域的方案。主论文中采用了基于半径的球查询方法。此处我们还实验了基于kNN的邻域搜索，并尝试了不同的搜索半径和k。本实验中，所有训练和测试均在均匀采样密度的ModelNet40形状上进行，使用1024个点。如表5所示，基于半径的球查询方法略优于基于kNN的方法。然而，我们推测在极不均匀的点集中，基于kNN的查询将导致较差的泛化能力。我们还观察到，稍大的半径有助于性能提升，可能是因为它能捕获更丰富的局部模式。

kNN (k=16)	kNN (k=64)	radius (r=0.1)	radius (r=0.2)
89.3	90.3	89.1	90.7

表5：邻域选择的影响。评估指标为ModelNet 40测试集上的分类准确率（%）。

C.3 最远点采样中随机性的影响。

在我们的集合抽象层级中，对于*Sampling layer*，我们使用最远点采样（FPS）进行点集子采样。然而，FPS算法具有随机性，子采样结果取决于首先选择的点。在此，我们评估模型对此随机性的敏感度。在表6中，我们测试了在ModelNet40上训练的模型的特征稳定性和分类稳定性。

To evaluate feature stability we extract global features of all test samples for 10 times with different random seed. Then we compute mean features for each shape across the 10 sampling. Then we compute standard deviation of the norms of feature's difference from the mean feature. At last we average all std. in all feature dimensions as reported in the table. Since features are normalized into 0 to 1 before processing, the 0.021 difference means a 2.1% deviation of feature norm.

For classification, we observe only a 0.17% standard deviation in test accuracy on all ModelNet40 test shapes, which is robust to sampling randomness.

Feature difference std.	Accuracy std.
0.021	0.0017

Table 6: Effects of randomness in FPS (using ModelNet40).

C.4 Time and Space Complexity.

Table 7 summarizes comparisons of time and space cost between a few point set based deep learning method. We record forward time with a batch size 8 using TensorFlow 1.1 with a single GTX 1080. The first batch is neglected since there is some preparation for GPU. While PointNet (vanilla) [20] has the best time efficiency, our model without density adaptive layers achieved smallest model size with fair speed.

It's worth noting that ours MSG, while it has good performance in non-uniformly sampled data, it's 2x expensive than SSG version due the multi-scale region feature extraction. Compared with MSG, MRG is more efficient since it uses regions across layers.

	PointNet (vanilla) 1	PointNet 1	Ours (SSG)	Ours (MSG)	Ours (MRG)
Model size (MB)	9.4	40	8.7	12	24
Forward time (ms)	11.6	25.3	82.4	163.2	87.0

Table 7: Model size and inference time (forward pass) of several networks.

为评估特征稳定性，我们使用不同随机种子对所有测试样本提取全局特征10次。随后计算每种形状在10次采样中的平均特征。接着计算特征与平均特征之差的范数的标准差。最后将所有特征维度的标准差取平均，结果如表所示。由于特征在处理前已归一化至0到1之间，0.021的差异意味着特征范数存在2.1%的偏差。

对于分类任务，我们在所有ModelNet40测试形状上仅观察到0.17%的标准差，这表明其对采样随机性具有鲁棒性。

Feature difference std.	Accuracy std.
0.021	0.0017

表6：FPS中随机性的影响（使用ModelNet40）。

C.4 时间与空间复杂度。

表7总结了基于点集的深度学习方法在时间和空间成本上的比较。我们使用TensorFlow 1.1在单块GTX 1080上以批量大小8记录了前向时间。由于GPU存在一些准备工作，第一批被忽略。虽然PointNet（vanilla）[20]具有最佳的时间效率，但我们的模型在没有密度自适应层的情况下以合理的速度实现了最小的模型尺寸。

值得注意的是，我们的MSG虽然在非均匀采样数据上表现良好，但由于多尺度区域特征提取，其成本是SSG版本的2倍。与MSG相比，MRG更为高效，因为它使用了跨层的区域。

	PointNet (vanilla) 1	PointNet 1	Ours (SSG)	Ours (MSG)	Ours (MRG)
Model size (MB)	9.4	40	8.7	12	24
Forward time (ms)	11.6	25.3	82.4	163.2	87.0

表7：若干网络的模型规模与推理时间（前向传播）。